TUSHAR THAKUR

Computer Engineer | AI & Robotics Scholar | VIT Pune

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EDUCATION

Vishwakarma Institute of Technology (VIT), Pune 2022 - Present

Bachelor of Technology (B.Tech) - Artificial Intelligence & Data Science

Focus: Computer Vision, Embedded Robotics, Neural Architectures, Inverse Kinematics

TECHNICAL REPERTOIRE & SKILLS

Languages:	Python, C, C++, JavaScript, TypeScript, SQL, HTML5, CSS3
AI & Vision:	PyTorch, OpenCV, YOLOv8, Computer Vision, Deep Learning, TensorRT, Scikit-Learn
Hardware & IoT:	ESP32 (FreeRTOS/ESP-IDF), Arduino, Raspberry Pi, Robotics Kinematics, KiCad
Frameworks & Tools:	React, Vite, Tailwind CSS, FastAPI, Flask, Git/GitHub, Docker, Linux
Design & Spatial:	Figma (Design Systems), Autodesk Fusion 360 (Parametric 3D CAD), Canva

KEY TECHNICAL PROJECTS

Synapse Vision AI (Autonomous Edge Object Detection & Spatial Tracking)

`Tech Stack: Python, PyTorch, OpenCV, TensorRT, ROS2`

- Architected an INT8 quantized multi-class computer vision pipeline sustaining 60 FPS on edge hardware.
- Implemented Kalman spatial trajectory prediction maintaining tracking through extreme camera occlusions.

Aegis Hexapod 18-DOF (Bio-inspired Omnidirectional Walker)

`Tech Stack: ESP32, C++, Autodesk Fusion 360, Inverse Kinematics`

- Modeled parametric 3-DOF leg assemblies and derived analytical closed-form inverse kinematics at 100Hz.
- Engineered dual-core FreeRTOS task segregation preventing power brown-outs across 18 high-torque servos.

Aether OS Interface (Contextual Minimalist Knowledge Operating System)

`Tech Stack: React, TypeScript, Tailwind CSS, Framer Motion, Vite`

- Designed and built an avant-garde digital dossier with responsive SVG botanical Bézier generative roots.
- Achieved 100/100 Lighthouse performance rating with sub-second page transitions.

EXPERIENCE & INSTITUTIONAL LEADERSHIP

Public Relations Officer & Design Coordinator - I2IAOC - VIT Pune 2023 - Present

Direct PR outreach and design direction for collegiate innovation summits with 3,500+ attendees.

Secretary - The Ideators Club - VIT Pune 2023 - Present

Lead operational governance and curate campus-wide hackathons yielding 28 functional prototypes.

Graphic Designer & R&D Assistant - The Robotics Forum - VIT Pune 2022 - 2023

Assisted core division in 3D CAD modeling, 3D printing chassis, and telemetry circuit debugging.